17 A disc is placed on horizontal ground.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The disc exerts a force of F newtons over an area of A cm^2 where

$F = 576.2$ correct to 1 decimal place
$A = 72$ correct to 2 significant figures.

The disc exerts a pressure of P newtons/cm^2 on the ground.

By considering bounds, calculate the value of P to a suitable degree of accuracy.
You must show all your working and give a reason for your answer.

$576.15 \leq F < 576.25$ N

1 mark
Half of 0.1 either side.

$71.5 \leq A < 72.5$ cm^2

1 mark
72 to 2sf: nearest unit.

Lower pressure = $\frac{576.15}{72.5}$

1 mark
Lower force ÷ upper area.

$= 7.946896551...$ N/cm^2

Upper pressure = $\frac{576.25}{71.5}$

1 mark
Both correct bounds.

$= 8.059440559...$ N/cm^2

Both bounds round to 8 to the nearest whole number.

Therefore $P = 8$ N/cm^2 to 1 significant figure.

1 mark
State precision and reason.

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(Total for Question 17 is 5 marks)

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